ESP32-Based Home Light Automation

Project Details, Hardware Connections, Blynk Setup, Troubleshooting & Complete Arduino Code

Project goal

Control an outdoor light remotely using an ESP32, a relay module, and the Blynk IoT app. The phone can use mobile data or another Wi-Fi network; the ESP32 itself must stay powered and connected to the home 2.4 GHz Wi-Fi and the internet.

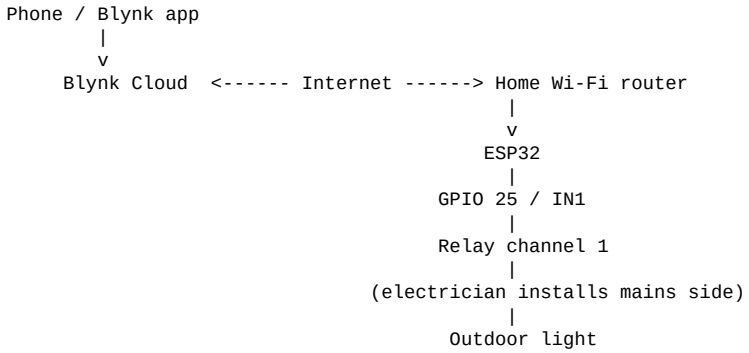
Important safety note

The street light supply was measured around 219–230 V AC. Mains voltage can cause fatal electric shock or fire. This guide covers only low-voltage ESP32/relay control wiring. A qualified electrician must install the mains-side COM/NO/NC wiring, enclosure, fuse/protection, insulation, and strain relief. Keep all mains contacts disconnected while testing.

Prepared as a practical beginner guide

1. How the Project Works

The Blynk app sends a switch state to Blynk Cloud. The ESP32 connects to that cloud service through the router. When Virtual Pin V0 becomes 1, the ESP32 drives GPIO 25 LOW to activate a typical active-LOW relay input. When V0 becomes 0, GPIO 25 goes HIGH to turn the relay off.



2. Hardware Required

Part	Purpose / notes
ESP32 DevKit (ESP32-D0WD-V3)	Controller; GPIO 25 is used for relay input.
5 V two-channel relay module	Switches the load using isolated relay contacts; check board labels and coil rating.
Regulated 5 V DC supply	Power for the ESP32 through its 5V/VIN input and/or relay module, as appropriate for the board.
Wi-Fi router with 2.4 GHz enabled	ESP32 uses 2.4 GHz Wi-Fi; internet access is required for Blynk Cloud.
Phone with Blynk IoT app	Remote switch interface using Virtual Pin V0.
Wires and protected enclosure	Keep low-voltage and mains wiring physically separated.

3. Low-Voltage Wiring (Test Only)

Use only the low-voltage side during initial tests. Disconnect mains wiring from COM/NO/NC. The board previously described has GND, IN1, IN2, VCC and a JD-VCC/VCC jumper; confirm labels on your actual board before wiring.

Relay/control module	Connection
IN1	ESP32 GPIO 25
IN2	Leave disconnected unless using a second relay channel.
GND	Common ground with ESP32 and the low-voltage supply, as required by the module design.
VCC	Regulated 5 V only if the module is marked for 5 V. Never apply 24 V to a 5 V relay board.
JD-VCC/VCC jumper	For a typical non-isolated setup, keep the supplied jumper installed. Follow the exact module's docur
COM / NO / NC	Leave disconnected during low-voltage testing. Mains-side installation is for a qualified electrician on

Power caution: Do not connect 24 V directly to ESP32 5V/VIN, 3V3, or a 5 V relay module. A 5 V AC/DC supply's mains input must be installed safely by a qualified person. Avoid connecting USB and an external supply simultaneously unless the board's power arrangement is confirmed safe.

4. Blynk IoT Setup

• Create or open the Blynk template named Home light automation.

• Create an Integer datastream on Virtual Pin V0, with values 0 and 1.

• Add a Switch widget to the dashboard and link it to V0.

• Create a device from the template and copy its Auth Token into the code locally.

• The ESP32 connects to the home router and Blynk Cloud. The phone may be on mobile data or another Wi-Fi network.

5. Complete Arduino Code

Replace the three placeholders with your own current Blynk token, Wi-Fi name, and Wi-Fi password. Do not share those values in screenshots or messages. Because a token and Wi-Fi password were pasted into the conversation, regenerate the Blynk token and change the Wi-Fi password before using this project.

```
#define BLYNK_TEMPLATE_ID "TMPL3-jWnnwP5"
#define BLYNK_TEMPLATE_NAME "Home light automation"
#define BLYNK_AUTH_TOKEN "YOUR_NEW_BLYNK_AUTH_TOKEN"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>

char ssid[] = "YOUR_WIFI_NAME";
char pass[] = "YOUR_WIFI_PASSWORD";

#define RELAY1 25

// Typical active-LOW relay input:
// LOW = ON, HIGH = OFF.
// Verify this logic for your exact relay module.

BLYNK_WRITE(V0)
{
  int value = param.asInt();

  if (value == 1) {
    digitalWrite(RELAY1, LOW);
    Serial.println("RELAY 1: ON");
  } else {
    digitalWrite(RELAY1, HIGH);
    Serial.println("RELAY 1: OFF");
  }
}

BLYNK_CONNECTED()
{
  Serial.println("Blynk Cloud connected");
  Blynk.syncVirtual(V0);
}

void setup()
{
  Serial.begin(115200);
  delay(500);

  pinMode(RELAY1, OUTPUT);
  digitalWrite(RELAY1, HIGH);  // Keep relay OFF at startup

  Serial.println();
  Serial.println("HOME LIGHT AUTOMATION");
  Serial.println("Starting Wi-Fi and Blynk...");

  WiFi.mode(WIFI_STA);
  WiFi.setSleep(false);

  // Blynk.begin handles initial Wi-Fi and cloud connection.
  // The main loop below services Blynk and reconnects if needed.
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  Serial.println("Initial Blynk connection complete");
}

void loop()
{
  if (WiFi.status() != WL_CONNECTED) {
    Serial.println("Wi-Fi lost; restarting connection...");
    WiFi.disconnect();
    WiFi.begin(ssid, pass);
    delay(1000);
    return;
```

```
  }

  if (!Blynk.connected()) {
    Serial.println("Blynk lost; reconnecting...");
    Blynk.connect(1000);
  }

  Blynk.run();
}
```

6. Upload and Test

1. Disconnect the relay and external supply. Connect only ESP32 to the PC USB cable for uploading.

2. In Arduino IDE select the correct ESP32 board and COM port. If upload errors occur, set Upload Speed to 115200, close Serial Monitor, and retry. Hold BOOT only if needed during Connecting.

3. Wait for a successful upload with no fatal error.

4. Test the ESP32 alone using Serial Monitor at 115200 baud. Check Wi-Fi and Blynk connection.

5. Power the ESP32 from a known-good regulated 5 V supply, with the relay still disconnected. Verify that it reconnects to Blynk after USB is removed.

6. Only after the ESP32 works from external power, reconnect the relay low-voltage wires one at a time with COM/NO/NC still disconnected.

7. If Wi-Fi fails when the relay's VCC/GND are connected even with IN1 removed, investigate the relay supply, common-ground arrangement, wiring, or faulty module. Do not keep trying to upload with suspect hardware attached.

7. Troubleshooting

Symptom	What to check
Upload works without relay, fails with relay	Disconnect all relay wires before uploading. The upload warning about flash communication means external w
Red ESP32 LED is ON but Wi-Fi is offline	Power LED only confirms some power is present; it does not prove the ESP32 has booted. Check regulated
Square symbols in Serial Monitor	Set Serial Monitor to 115200 baud for the sketch. Bootloader messages may briefly appear at another baud ra
Blynk switch does not control relay	Check V0 datastream and Switch widget, correct device/token, GPIO25-to-IN1 wiring, common ground, 5 V re
Wi-Fi drops when relay is powered	Keep mains contacts disconnected. Test relay power without IN1. Use a suitable regulated supply and inspect

8. Safety Checklist

• Keep all 230 V AC wiring disconnected during ESP32 and relay testing.

• A qualified electrician must install the mains side, suitable enclosure, fusing/protection, insulation, and cable strain relief.

• Never connect mains voltage to ESP32 GPIO, 3V3, 5V/VIN, or relay coil pins.

• Never apply 24 V directly to a 5 V relay module or ESP32 power input.

• Do not rely on a software state alone as a safety isolation device. Use suitable hardware protection and a fail-safe installation.

• The code initializes the relay OFF (HIGH), but confirm the actual relay logic and startup behavior on your module before connecting a real load.

Project summary: ESP32 GPIO25 controls relay channel 1; Blynk Virtual Pin V0 controls the switch. USB is required for uploading, not for normal operation, provided the ESP32 has a reliable independent 5 V supply and working Wi-Fi/internet.

Security note: Never publish your Blynk Auth Token or Wi-Fi password. Replace exposed credentials before deploying the system.